



SCEGGS Darlinghurst

2008

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

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Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks)

(a) Find $\int \frac{dx}{\sqrt{16x^2 - 1}}$ 2

(b) Evaluate $\int_1^e x \ln x \, dx$ 3

(c) (i) Find real numbers a and b such that 2

$$\frac{5x^2 + x + 8}{(x+1)(x^2 + 3)} \equiv \frac{a}{x+1} + \frac{bx-1}{x^2 + 3}$$

(ii) Hence find $\int \frac{5x^2 + x + 8}{(x+1)(x^2 + 3)} \, dx$ 2

(d) Find $\int \tan^3 x \, dx$ 2

(e) Using a suitable substitution, or otherwise, evaluate: 4

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} \, dx$$

End of Question 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) Let $\alpha = 1 - \sqrt{3}i$.

(i) Find the exact value of $|\alpha|$ and $\arg \alpha$. **2**

(ii) Hence express $(1 - \sqrt{3}i)^{10}$ in modulus-argument form. **1**

(b) Express $\sqrt{7 - 24i}$ in the form $a + ib$, where a and b are real. **3**

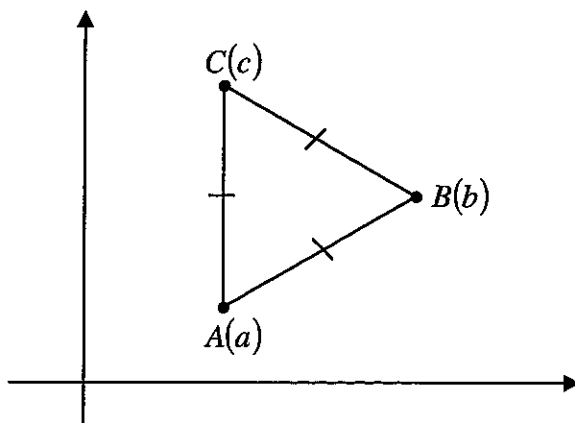
(c) Sketch the region in the complex plane where the two inequalities **3**
 $0 \leq \text{Arg}(z) \leq \frac{3\pi}{4}$ and $|z - 2i| \geq |z|$ both hold.

(d) Sketch the locus of z satisfying $|z - 3| + |z + 3| = 10$. **3**
Show any intercepts with the axes.

Question 2 continues on page 4

Question 2 (continued)

(e)



The points A , B and C on the Argand diagram represent the complex numbers a , b and c respectively. $\triangle ABC$ is equilateral.

Let $w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

(i) Show that $\frac{a-b}{c-b} = w$.

1

(ii) By writing another similar expression for w , prove that

2

$$a^2 + b^2 + c^2 = ab + bc + ca$$

End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 + 3x^2 - 5x - 2 = 0$ has roots α , β and γ . 2

Find a cubic equation with integer coefficients whose roots are $\frac{2}{\alpha}$, $\frac{2}{\beta}$ and $\frac{2}{\gamma}$.

- (b) Consider the curve $x^2 + y^2 + xy = 3$.

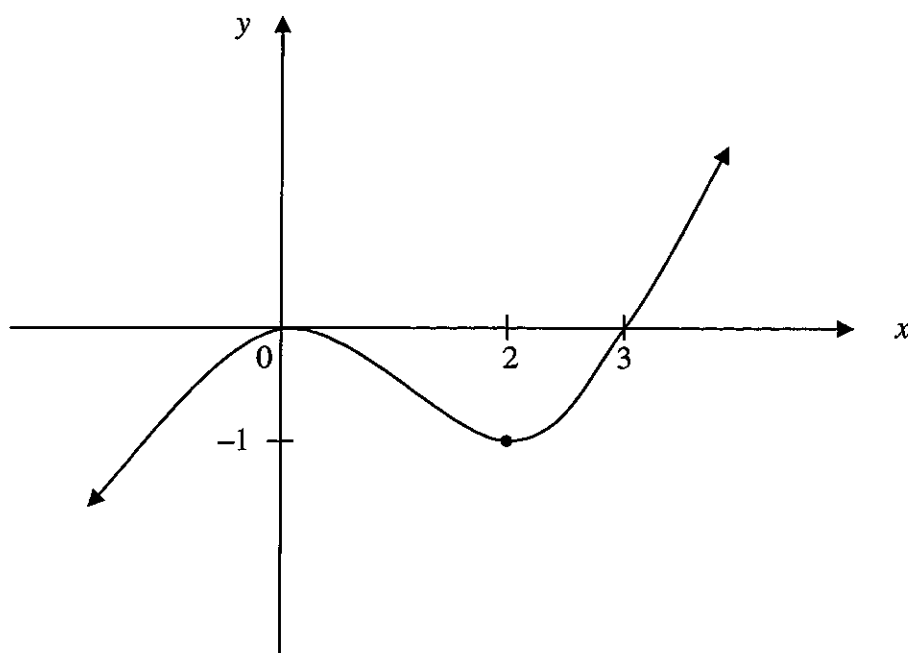
- (i) Show that $\frac{dy}{dx} = -\left(\frac{2x + y}{x + 2y}\right)$. 1

- (ii) Hence find the coordinates of any stationary points. 2

Question 3 continues on page 6

Question 3 (continued)

- (c) The diagram shows the graph of $y = f(x)$ where $f(x) = \frac{1}{4}x^2(x-3)$.



On the answer page provided, draw separate sketches of the graphs of the following:

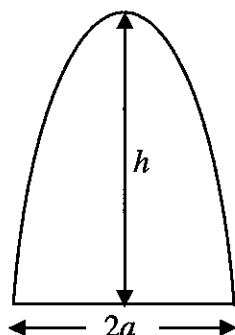
- | | | |
|-------|----------------------------|----------|
| (i) | $y = \frac{1}{4}x^2 x-3 $ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 1 |
| (iii) | $y^2 = -f(x)$ | 2 |
| (iv) | $y = \tan^{-1}(f(x))$ | 2 |

Question 3 continues on page 7

Question 3 (continued)

(d) (i)

1

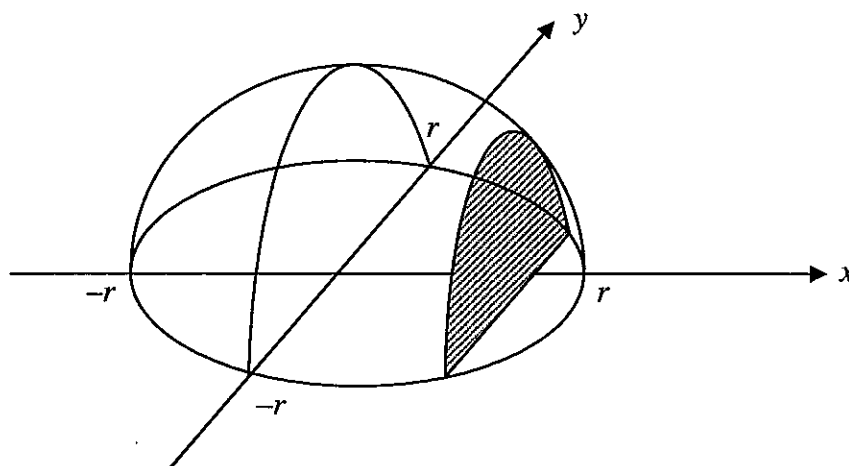


A parabolic segment has height h and width $2a$.

Use Simpson's Rule with three function values, to show that the exact area of this segment is $\frac{4ah}{3}$.

(ii)

3



The base of a solid is the region in the xy plane enclosed by the circle $x^2 + y^2 = r^2$.

Each cross-section perpendicular to the x -axis is a parabolic segment with height one half its width.

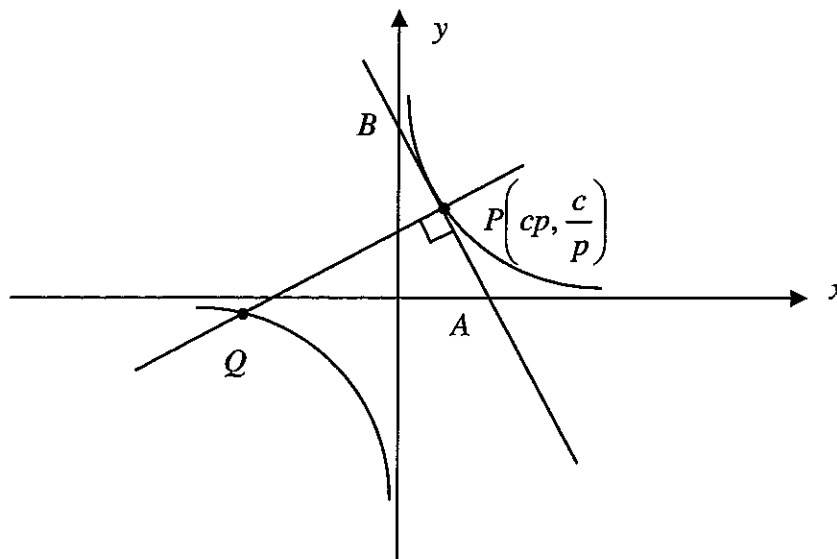
Show that the volume of the solid is $\frac{16r^3}{9}$ units³.

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) The point $P\left(cp, \frac{c}{p}\right)$ is a point on the hyperbola $xy = c^2$.

The tangent to the hyperbola at P intersects the x and y axes at A and B respectively and the normal to the hyperbola at P intersects the second branch at Q .



- (i) Show that the equation of the normal at P is $py - c = p^3(x - cp)$. 2

- (ii) Show that the x coordinates of P and Q satisfy the equation 2

$$x^2 - c\left(p - \frac{1}{p^3}\right)x - \frac{c^2}{p^2} = 0$$

and hence find the coordinates of Q .

- (iii) Given the distance $AB = 2c\sqrt{p^2 + \frac{1}{p^2}}$, show that the 2

$$\text{area of } \triangle ABQ = c^2\left(p^2 + \frac{1}{p^2}\right)^2.$$

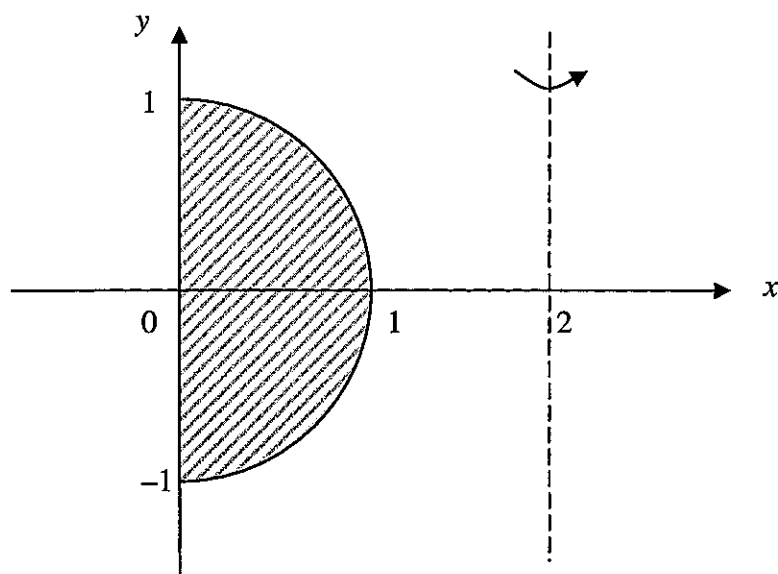
- (iv) Find the minimum area of $\triangle ABQ$. 1

(You may use the inequality $\frac{a}{b} + \frac{b}{a} \geq 2$ for $a, b > 0$.)

Question 4 continues on page 9

Question 4 (continued)

(b)



The shaded semicircle in the diagram above is rotated about the line $x = 2$.

- (i) Using the method of cylindrical shells, show that the volume V of the resulting solid is given by 1

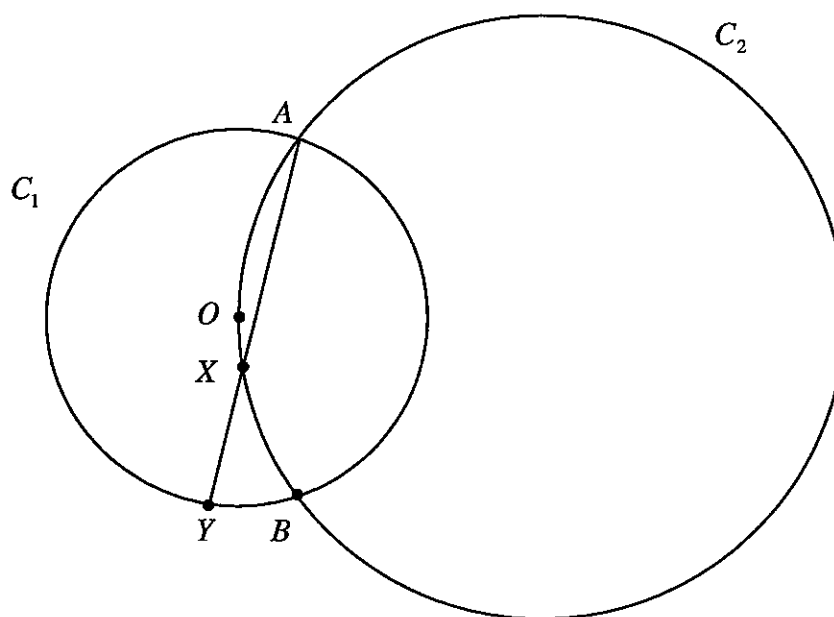
$$V = \int_0^1 4\pi (2 - x) \sqrt{1 - x^2} \, dx$$

- (ii) Hence find the volume of the solid. 3

Question 4 continues on page 10

Question 4 (continued)

(c)



Two circles C_1 and C_2 intersect at A and B . C_2 passes through O , the centre of C_1 . X lies on the arc AOB and AX intersects C_1 again at Y .

- (i) State why $\angle AOB = 2 \times \angle AYB$. 1
- (ii) Prove that $XY = XB$. 3

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that if α is a double root of $f(x) = 0$ then $f(\alpha) = f'(\alpha) = 0$. 2
- (ii) Find all roots of the equation $2x^3 - 5x^2 - 4x + 12 = 0$ given that two of the roots are equal. 3
- (b) (i) By drawing a diagram, or otherwise, find the solutions of $z^5 = 1$. 2
- (ii) Show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. 2
- (iii) Hence find the exact value of $\cos \frac{2\pi}{5}$. 2
- (c) 11 persons gather to play basketball by forming 2 teams of 5 to play each other. The remaining person acts as a referee.
- (i) In how many ways can the teams be formed? 2
- (ii) If two particular persons are not to be in the same team, how many ways are there then to choose the teams? 2

End of Question 5

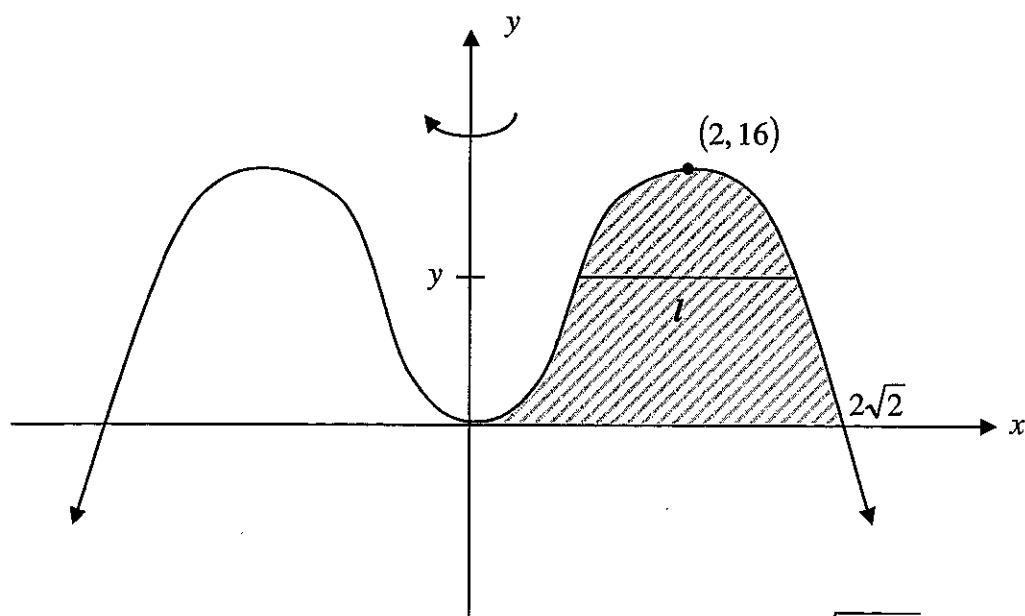
Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) The sequence $\{a_n\}$ is given by: 3

$$a_1 = 2, a_2 = \frac{3}{2} \text{ and } (n+1)a_{n+1} = a_{n-1} - (n-2)a_n \text{ for } n > 1.$$

Prove by induction that for $n \geq 1$, $a_n = \frac{n+1}{n!}$

- (b) The region bound by the curve $y = 8x^2 - x^4$ and the x axis in the first quadrant is rotated about the y axis to form a solid. When the region is rotated, the horizontal line segment l at height y sweeps out an annulus.



- (i) Show that the area of the annulus at height y is given by $2\pi\sqrt{16-y}$. 3

- (ii) Find the volume of the solid. 2

Question 6 continues on page 13

Question 6 (continued)

- (c) (i) Differentiate $x \cos^{n-1} x$. **1**

- (ii) Let $I_n = \int_0^{\frac{\pi}{2}} x \cos^n x \, dx$ for $n = 0, 1, 2, \dots$ **4**

Show that for $n \geq 2$

$$I_n = -\frac{1}{n^2} + \frac{n-1}{n} I_{n-2}$$

- (iii) Hence evaluate $\int_0^{\frac{\pi}{2}} x \cos^4 x \, dx$. **2**

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) If $z = \cos \theta + i \sin \theta$, show that $z + \frac{1}{z} = 2 \cos \theta$ and 2

$$z^n + \frac{1}{z^n} = 2 \cos n\theta.$$

- (ii) Hence show that $\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$. 2

- (iii) Hence find the general solution to the equation 3

$$16 \cos^5 \theta = 15 \cos 3\theta + \cos 5\theta.$$

Question 7 continues on page 15

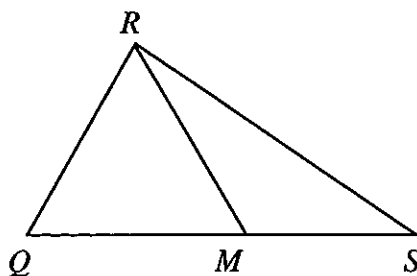
Question 7 (continued)

For parts (b) and (c) you may use the following identity:

$$\text{If } \frac{P}{Q} = \frac{R}{S}, \text{ then } \frac{P}{Q} = \frac{R}{S} = \frac{P \pm R}{Q \pm S}$$

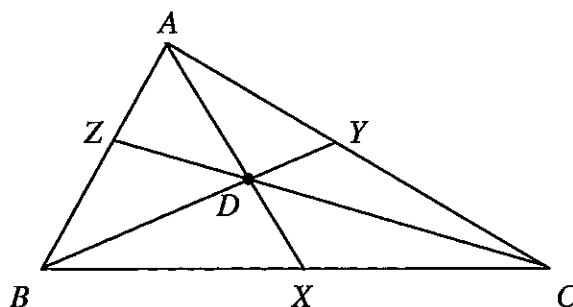
(b) (i)

1



$$\text{Show that } \frac{QM}{MS} = \frac{\text{Area } \triangle RQM}{\text{Area } \triangle RMS}.$$

(ii)



In the diagram, Z, X and Y lie on the sides of $\triangle ABC$ AB, BC and CA respectively such that AX, BY and CZ are concurrent. D is the point of concurrency.

$$(\alpha) \text{ Show that } \frac{BX}{XC} = \frac{\text{Area } \triangle ABD}{\text{Area } \triangle ACD}. \quad 2$$

$$(\beta) \text{ Hence prove } \frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = 1. \quad 2$$

Question 7 continues on page 16

Question 7 (continued)

(c) a, x, y, z are real numbers such that

$$\frac{\cos x + \cos y + \cos z}{\cos (x + y + z)} = \frac{\sin x + \sin y + \sin z}{\sin (x + y + z)} = a$$

(i) Use the identity given earlier to show that 1

$$a = \frac{cisx + cisy + cisz}{cis (x + y + z)}$$

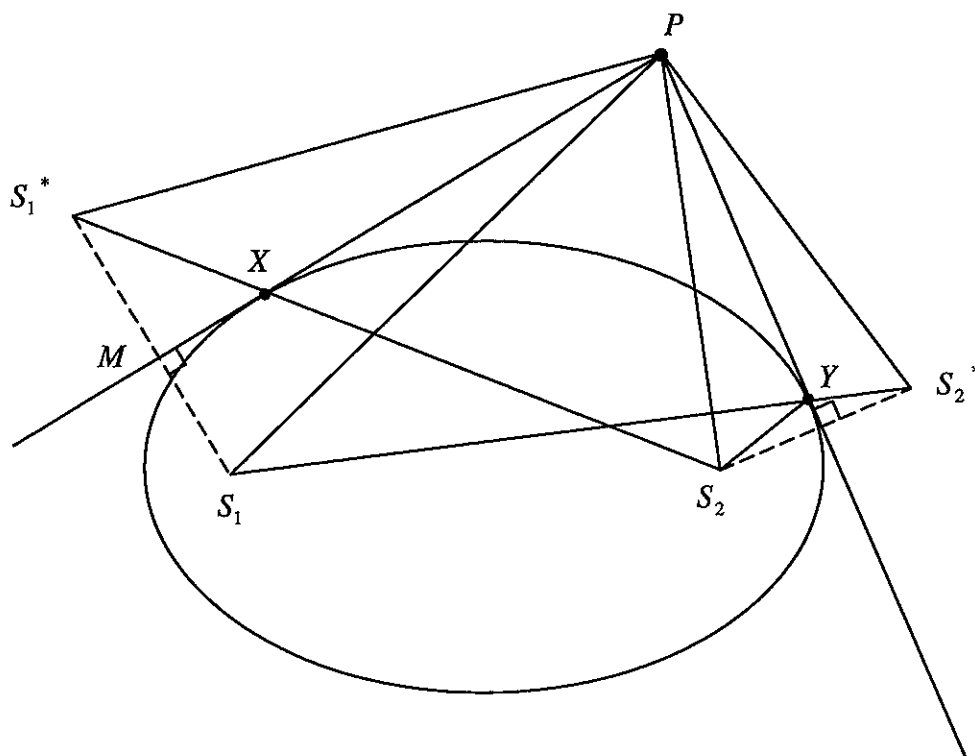
(ii) Hence show that 2

$$a = \cos(y + z) + \cos(x + z) + \cos(x + y)$$

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet.

(a)



In the diagram, X and Y are arbitrary points on the ellipse and tangents to the ellipse at X and Y meet at the point P . The points S_1 and S_2 are the foci of the ellipse, and S_1^* and S_2^* are the reflections of S_1 and S_2 across the tangents, as shown. $S_1S_1^*$ and the tangent at X intersect at the point M .

You may assume, without proof, the following two properties of an ellipse:

1. The sum of the focal lengths from any point on an ellipse is constant.
2. The reflection property:
Tangents to an ellipse are equally inclined to the focal chords drawn through the point of contact.

- (i) Prove $\triangle MXS_1 \equiv \triangle MXS_1^*$ and hence show that $S_1^*XS_2$ is a straight line. [Note that similarly, $S_1YS_2^*$ is a straight line.] 3
- (ii) Prove that $S_1^*S_2 = S_1S_2^*$ 2
- (iii) Hence state why $\triangle S_1^*PS_2 \equiv \triangle S_1PS_2^*$. 1
- (iv) Deduce that $\angle S_1PX = \angle S_2PY$. 2

Question 8 continues on page 18

Question 8 (continued)

(b) (i) What value of x maximizes the expression $\log_e x - x + 1$? 1

(ii) Deduce that $\log_e x \leq x - 1$ for $x > 0$. 1

(iii) Consider the set of n positive numbers 2

$$p_1, p_2, \dots, p_n \text{ such that } p_1 + p_2 + \dots + p_n = 1.$$

Use the result in part (ii) to show that

$$\log_e(np_1) + \log_e(np_2) + \dots + \log_e(np_n) \leq 0$$

(iv) Deduce that $n^n p_1 p_2 \dots p_n \leq 1$. 1

(v) Let $A = x_1 + x_2 + \dots + x_n$ ($x_1, x_2, \dots, x_n \geq 0$) and set 1

$$p_1 = \frac{x_1}{A}, p_2 = \frac{x_2}{A}, \dots, p_n = \frac{x_n}{A}.$$

$$\text{Prove that } \frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}.$$

(vi) Show that for $a, b, c, d > 0$, with $abcd = 1$ 1
 $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd \geq 10.$

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

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Mathematics Extension 2

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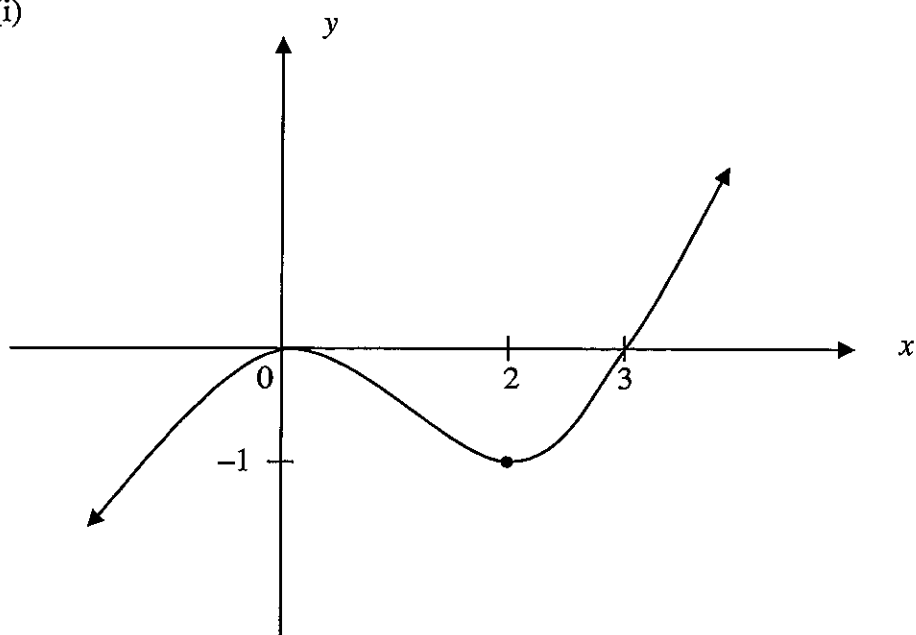
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Questions 3 (c)

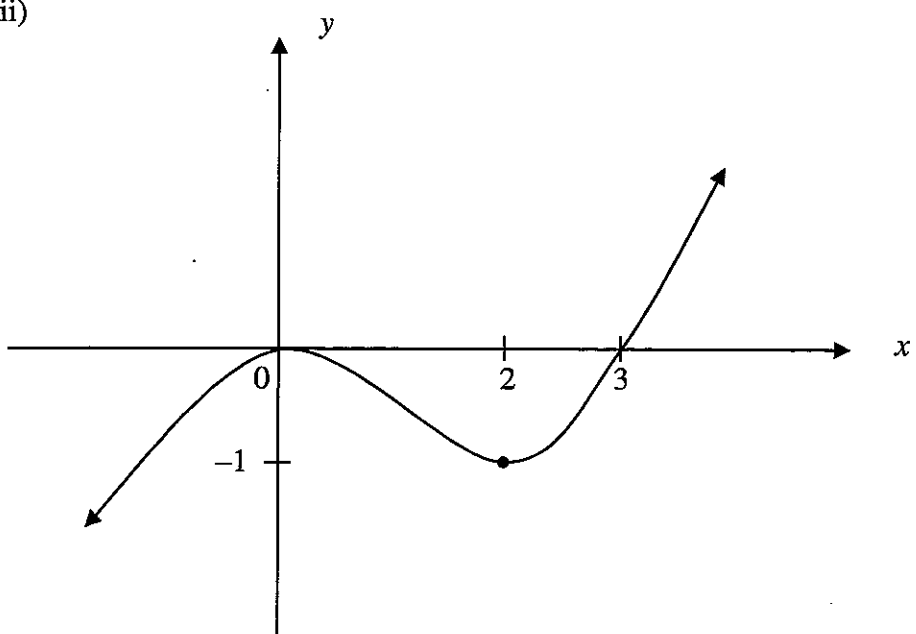
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Student Number

(i)



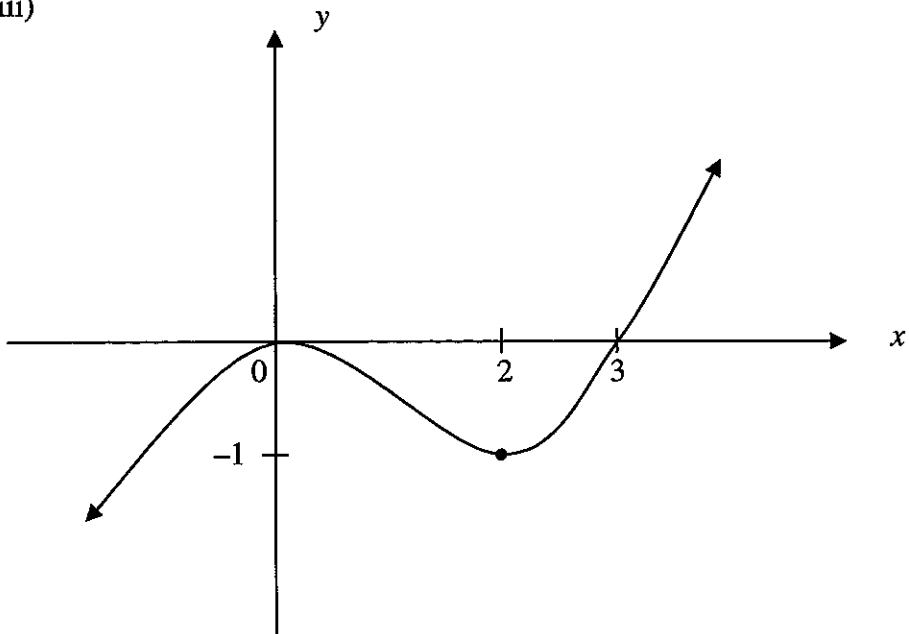
(ii)



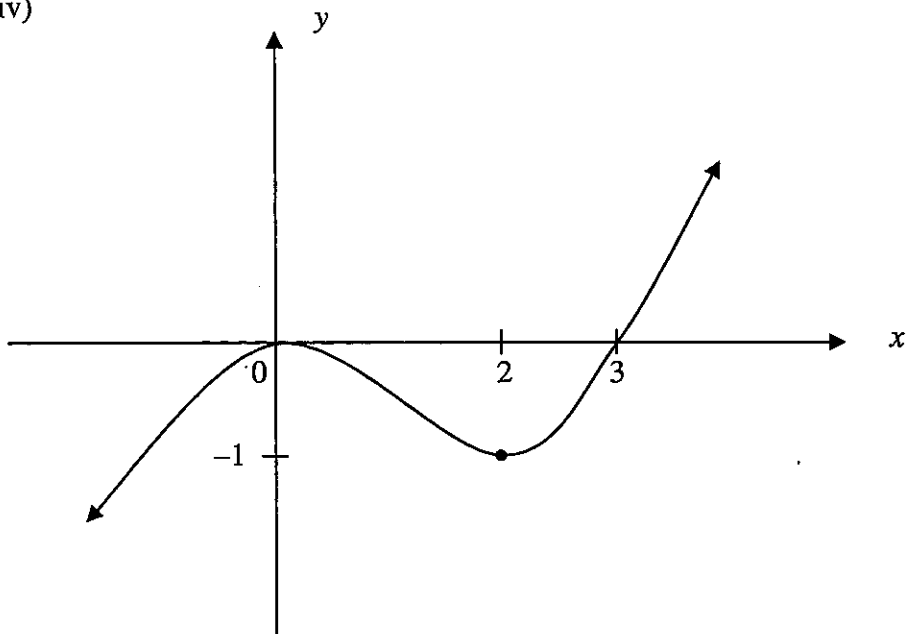
Question 3(c) continues on next page

Question 3 (continued)

(c) (iii)



(iv)



Question 1 (15 marks)

Calc /15

(a) $\int \frac{dx}{\sqrt{16x^2-1}}$

$$= \frac{1}{4} \ln(4x + \sqrt{16x^2-1}) + C$$

Reverse chain rule !!!

(b) $u = \ln x$ $dv = x$
 $du = \frac{1}{x}$ $v = \frac{x^2}{2}$

$$\int x \ln x \, dx$$

$$= \left[\frac{x^2 \ln x}{2} \right]_1^e - \int_1^e \frac{x}{2} \, dx \quad \checkmark$$

$$= \frac{e^2}{2} - \left[\frac{x^2}{4} \right]_1^e \quad \checkmark$$

$$= \frac{e^2}{2} - \left(\frac{e^2}{4} - \frac{1}{4} \right)$$

$$= \frac{e^2}{4} + \frac{1}{4} \quad \checkmark$$

(c) (i) $a = 3$ $\checkmark$
 $b = 2$ $\checkmark$

(ii) $\int \frac{5x^2 + x + 8}{(x+1)(x^2+3)} \, dx$

$$= \int \frac{3}{x+1} + \frac{2x-1}{x^2+3} \, dx$$

$$= 3 \ln(x+1) + \ln(x^2+3) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) \quad \checkmark$$

(d) $\int \tan^3 x \, dx$

Unfortunately some silly mistakes here

$$= \int \tan x (\sec^2 x - 1) \, dx$$

eg. $\tan^2 x \neq \sec^2 x + 1$

$$= \int \sec^2 x \tan x - \frac{\sin x}{\cos x} \, dx$$

$$= \frac{\tan^2 x}{2} + \ln(\cos x) + C \quad \checkmark$$

Q1 cont.

$$(e) \int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx$$

$$= \int_0^2 \frac{-(4-x^2)}{\sqrt{4-x^2}} + \frac{4}{\sqrt{4-x^2}} dx \checkmark$$

$$= \int_0^2 -\sqrt{4-x^2} + \left[4 \cdot \sin^{-1}\left(\frac{x}{2}\right) \right]_0^2 \checkmark$$

$$= -\frac{\pi \cdot 2^2}{4} + \left[4 \cdot \frac{\pi}{2} - 0 \right] \checkmark$$

$$= \pi$$

Whilst this is the easier method, no one saw to do it this way.

OR let $x = 2\sin\theta$
 $dx = 2\cos\theta d\theta$
 $x=0 \rightarrow \theta=0$
 $x=2 \rightarrow \theta=\pi/2 \checkmark$

This should have been an easy substitution to spot

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx$$

$$= \int_0^{\pi/2} \frac{4\sin^2\theta}{2\cos\theta} 2\cos\theta d\theta \checkmark$$

$$= \int_0^{\pi/2} 4\sin^2\theta d\theta$$

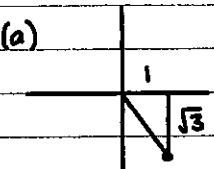
$$= \int_0^{\pi/2} 2(1-\cos 2\theta) d\theta \checkmark$$

$$= \left[2\theta - \sin 2\theta \right]_0^{\pi/2}$$

$$= \pi \checkmark$$

Question 2 (15 marks)

Comm 16, Reas 13

(a)  (i) $|a| = 2 \checkmark$
 $\arg a = -\pi/3 \checkmark$

(ii) $(1-\sqrt{3}i)^{10} = (2\text{cis}^{-\pi/3})^{10}$
 $= 2^{10}\text{cis}(-10\pi/3)$
 $= 2^{10}\text{cis}(2\pi/3) \checkmark$

Always simplify the argument to the Principle Argument!

(b) $a+ib = \sqrt{7-24i}$
 $(a+ib)^2 = 7-24i$
 $(a^2-b^2) + 2abi = 7-24i \checkmark$
 $a^2-b^2 = 7 \quad \textcircled{1}$
 $2ab = -24 \quad \textcircled{2}$

It would have been terribly mean of me to have given a quadratic with irrational roots! If your answer doesn't seem correct perhaps you should check over your working.

$\textcircled{2} \rightarrow b = -\frac{12}{a}$

sub in $\textcircled{1} \rightarrow a^2 - \frac{144}{a^2} = 7$

$$a^4 - 7a^2 - 144 = 0$$

$$(a^2-16)(a^2+9) = 0$$

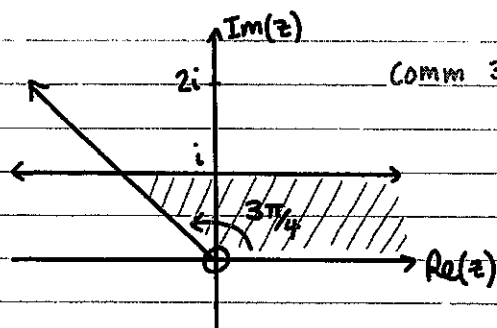
$$a = +4, -4 \text{ (since } a \in \mathbb{R}) \checkmark$$

$$b = -3, +3$$

$$\sqrt{7-24i} = 4-3i, -4+3i \checkmark$$

Q2 cont.

(c)



Comm 3

- ✓ $0 \leq \text{Arg}(z) \leq 3\pi/4$
- ✓ $|z-2i| = |z|$
- ✓ correct region

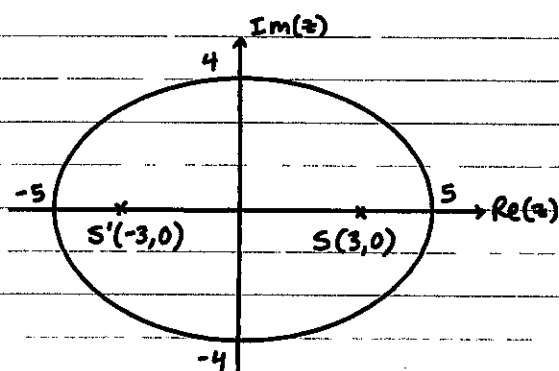
You should be able to sketch $|z-2i| \geq |z|$ by just thinking about. You shouldn't have to resort to algebra

(d) $|z-3| + |z+3| = 10$

Ellipse $\rightarrow$ Foci $(\pm 3, 0)$ ie. $ae = 3$
 $\rightarrow 2a = 10$

$\therefore a = 5, e = 3/5$

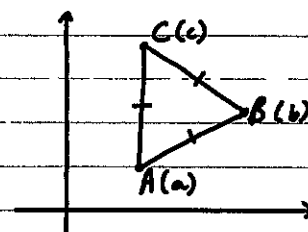
$$\begin{aligned} b^2 &= a^2(1-e^2) \\ &= 25(1-9/25) \\ &= 16 \\ b &= 4 \end{aligned}$$



Comm 3

Q2 cont.

(e)



Reas 3

(i) $\vec{BA} = \vec{BC} \times \text{cis } \pi/3$
 (anticlockwise rotation by $\pi/3$)
 $a-b = (c-b)\omega$
 $\frac{a-b}{c-b} = \omega$ ✓

(ii) Similarly $\frac{c-a}{b-a} = \omega$ ✓ $\frac{b-c}{a-c} = \omega$ also works

$$\Rightarrow \frac{a-b}{c-b} = \frac{c-a}{b-a}$$

$$(a-b)(b-a) = (c-a)(c-b)$$

$$ab - a^2 - b^2 + ab = c^2 - bc - ac + ab$$

$$ab + bc + ca = a^2 + b^2 + c^2$$
 ✓

Never skip a question without even trying, even if it's the style of question you dread most. It might turn out to be quite easy.

Question 3 (15 marks)

Comm /6

(a) $x^3 + 3x^2 - 5x - 2 = 0$ has roots α, β, γ

let $y = \frac{2}{x}$ (ie $x = \frac{2}{y}$)

for equation with roots $\frac{2}{\alpha}, \frac{2}{\beta}, \frac{2}{\gamma}$

$$\left(\frac{2}{y}\right)^3 + 3\left(\frac{2}{y}\right)^2 - 5\left(\frac{2}{y}\right) - 2 = 0 \quad \checkmark$$

$$8 + 12y - 10y^2 - 2y^3 = 0$$

$$y^3 + 5y^2 - 6y + 4 = 0 \quad \checkmark$$

(b) $x^2 + y^2 + xy = 3$

(i) $2x + 2y \cdot \frac{dy}{dx} + \left(x \cdot \frac{dy}{dx} + y\right) = 0$

$$\frac{dy}{dx} (x + 2y) = -2x - y$$

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y} \quad \checkmark$$

(ii) For S.P. $\frac{dy}{dx} = 0$

$$y = -2x$$

sub. into eqn:

$$x^2 + (-2x)^2 + x(-2x) = 3$$

$$3x^2 = 3$$

$$x = +1, -1 \quad \checkmark$$

$$y = -2, +2$$

$$\therefore \text{SP: } (1, -2), (-1, +2) \quad \checkmark$$

Substitute back into $y = -2x$ to obtain y coord.
The original equation gives 4 points — not all of which are stationary.

Comm 6

2008 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION
Mathematics Extension 2

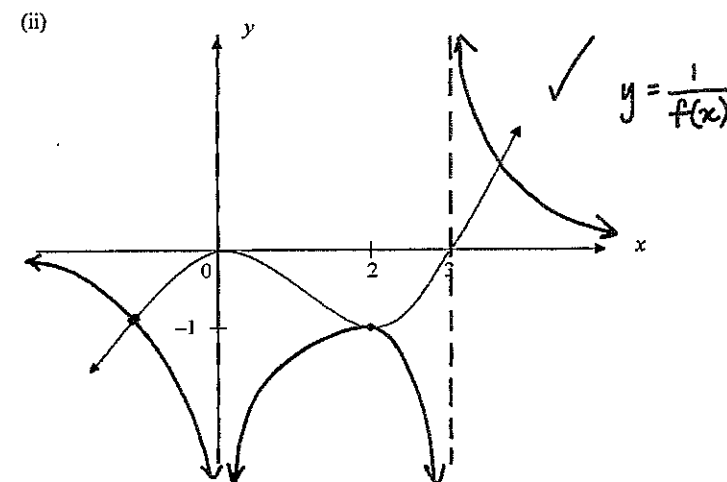
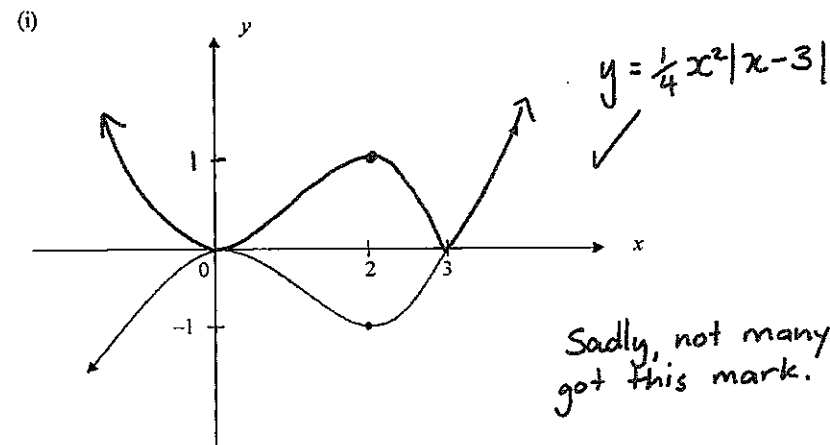
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Questions 3 (c)

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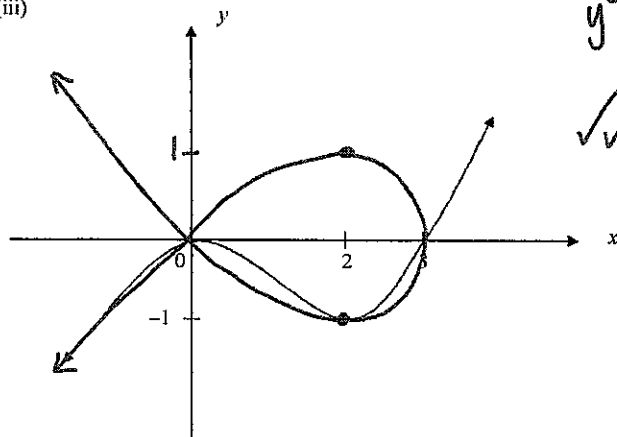
Student Number



Question 3(c) continues on next page

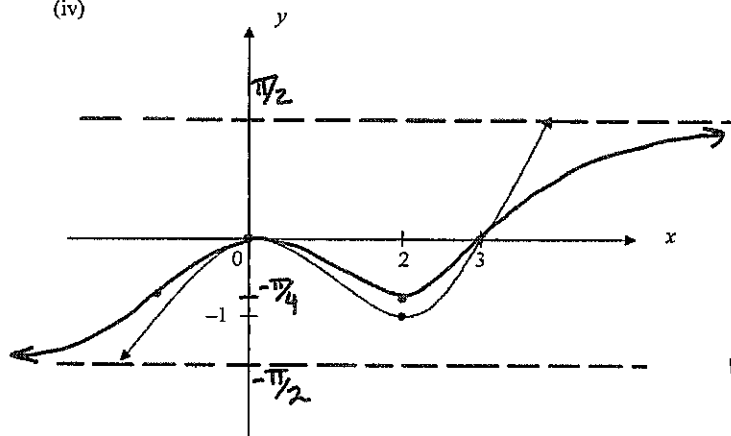
Question 3 (continued)

(c) (iii)



$$y^2 = -f(x)$$

(iv)



Not many graphs
had asymptotes!

Q3 cont.

(d) (i) Area of Parabolic Segment

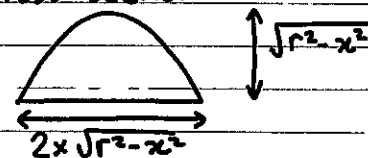
$$= \frac{h}{3} (y_1 + 4y_2 + y_3)$$

$$= \frac{a}{3} (0 + 4 \times h + 0)$$

$$= \frac{4ah}{3}$$

Simpson's rule approximates areas by finding the area bound by the parabola through 3 given points. Thus, when used to find the area bound by a parabola it is actually EXACT.

(ii) Cross-section:



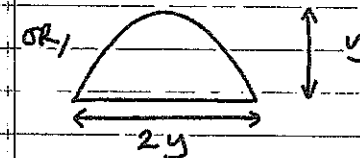
$$\text{Area} = \frac{4ah}{3} = \frac{4(r^2 - x^2)}{3}$$

$$\text{Volume} = 2 \times \int_0^r \frac{4(r^2 - x^2)}{3} dx \checkmark$$

$$= \frac{8}{3} \left[r^2 x - \frac{x^3}{3} \right]_0^r \checkmark$$

$$= \frac{8}{3} \left[\frac{2r^3}{3} \right]$$

$$= \frac{16r^3}{9} \text{ units}^3 \checkmark$$



$$\text{Area} = \frac{4ah}{3} = \frac{4y^2}{3} = \frac{4(r^2 - x^2)}{3}$$

Question 4 (15 marks)

Calc /4 Reas /5

$$\begin{aligned} \text{a) (i) } x &= cp & y &= \frac{c}{p} \\ \frac{dx}{dp} &= c & \frac{dy}{dp} &= -\frac{c}{p^2} \\ \frac{dy}{dx} &= \frac{-\frac{c}{p^2}}{c} = -\frac{1}{p^2} \quad \checkmark \\ m_N &= p^2 \end{aligned}$$

EQU NORMAL:

$$\begin{aligned} y - \frac{c}{p} &= p^2(x - cp) \\ py - c &= p^3(x - cp) \quad \checkmark \end{aligned}$$

(ii) For points of int. P & Q solve simult.

$$\begin{aligned} y &= \frac{c^2}{x} \quad \& \quad py - c = p^3(x - cp) \\ \Rightarrow p\left(\frac{c^2}{x}\right) - c &= p^3x - cp^4 \end{aligned}$$

$$\Rightarrow p^3x^2 - cp^4x + c^2x - pc^2 = 0$$

$$\Rightarrow x^2 - c\left(p - \frac{1}{p^3}\right)x - \frac{c^2}{p^2} = 0 \quad \checkmark$$

Since x coord of P(cp) & Q(?) are the roots of this equation &

Product of roots = $-\frac{c^2}{p^2}$

$$\Rightarrow x \text{ coord } Q = -\frac{c}{p^3}$$

$$\therefore y \text{ coord } Q = \frac{c^2}{x} = -cp^3$$

$$Q\left(-\frac{c}{p^3}, -cp^3\right) \quad \checkmark$$

Calc 1

You can't show the x coord of P & Q satisfy the equation by substituting because you don't even know the coord of Q yet.

So you shouldn't even bother starting.

Q4 cont.

$$\text{(iii) } AB = 2c \sqrt{p^2 + \frac{1}{p^2}}$$

There was some poor fudging going on here.

$$\begin{aligned} PQ &= \sqrt{\left(cp + \frac{c}{p^3}\right)^2 + \left(\frac{c}{p} + cp^3\right)^2} \quad \checkmark \\ &= c \sqrt{p^2 + \frac{2}{p^2} + \frac{1}{p^6} + \frac{1}{p^2} + 2p^2 + p^6} \\ &= c \sqrt{p^6 + 3p^2 + \frac{3}{p^2} + \frac{1}{p^6}} \\ &= c \sqrt{\left(p^2 + \frac{1}{p^2}\right)^3} \end{aligned}$$

$$\text{Area } \triangle ABQ = \frac{1}{2} \times AB \times PQ$$

$$\begin{aligned} &= \frac{1}{2} \times 2c \sqrt{p^2 + \frac{1}{p^2}} \times c \sqrt{\left(p^2 + \frac{1}{p^2}\right)^3} \\ &= c^2 \left(p^2 + \frac{1}{p^2}\right)^2 \quad \checkmark \end{aligned}$$

$$\text{(iv) } \frac{a}{b} + \frac{b}{a} \geq 2 \quad \text{for } a, b > 0 \quad \text{Reas 1}$$

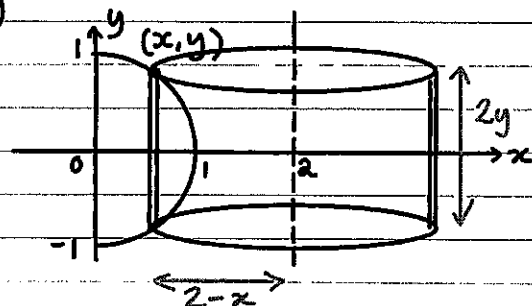
$$\Rightarrow p^2 + \frac{1}{p^2} \geq 2$$

$$\therefore \text{Minimum area} = 4c^2 \quad \checkmark$$

An easy mark if you understand inequalities & used the hint.

Q4 cont.

(b) (i)



Radius = $2-x$ Height = $2y$

$$\delta V = 2\pi r h$$

$$= 2\pi(2-x)2y$$

$$= 4\pi(2-x)\sqrt{1-x^2}$$

$$\therefore V = \int_0^1 4\pi(2-x)\sqrt{1-x^2} dx \quad \checkmark$$

(ii) $V = 8\pi \int_0^1 \sqrt{1-x^2} dx + 2\pi \int_0^1 -2x\sqrt{1-x^2} dx$ $\int_0^1 \sqrt{1-x^2} dx$ is $\frac{1}{4}$ of circle (not $\frac{1}{2}$)

Area of $\frac{1}{4}$ circle $\checkmark$

$$= 8\pi \times \frac{\pi 1^2}{4} + 2\pi \left[\frac{2}{3}(1-x^2)^{3/2} \right]_0^1 \quad \checkmark$$

$$= 2\pi^2 + 2\pi \left[0 - \frac{2}{3} \right]$$

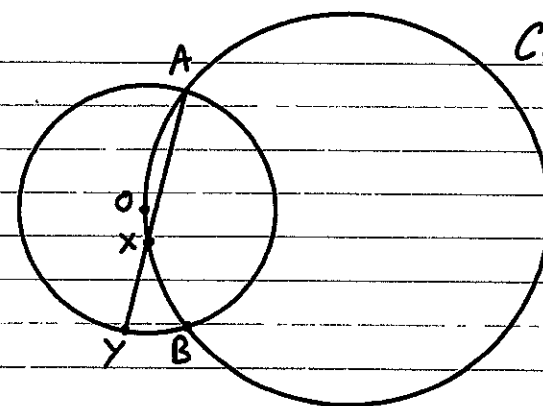
$$= 2\pi^2 - \frac{4\pi}{3} \text{ units}^3 \quad \checkmark$$

Calc 3

Q4 cont.

(c)

C_1



C_2 Reas 4

This is a really nice, straightforward question that could be written out clearly & efficiently

(i) Angle subtended at the centre is twice the angle subtended at the circumference, by the same arc $\checkmark$

(ii) let $\angle XYB = \alpha$

$\angle AOB = 2\alpha$ (from i)

$$\angle AXB = \angle AOB = 2\alpha$$

(angles in the same segment are equal) $\checkmark$

$$\angle XBY = \angle AXB - \angle XYB$$

$$= 2\alpha - \alpha$$

$$= \alpha$$

(exterior angle of Δ equals the sum of the 2 opposite interior angles) $\checkmark$

$$\therefore XY = XB$$

(sides opposite equal angles in a Δ are equal) $\checkmark$

Question 5 (15 marks)

Reas /8

(a)(i) Assume α is a double root of $f(x)=0$
then $f(x) = (x-\alpha)^2 Q(x)$ [& $f(\alpha)=0$]

The syllabus certainly states that you need to know this proof.

$$f'(x) = (x-\alpha)^2 Q'(x) + Q(x) \cdot 2(x-\alpha)$$

$$= (x-\alpha) [(x-\alpha)Q'(x) + 2Q(x)]$$

$$\therefore f'(\alpha) = (\alpha-\alpha) [(x-\alpha)Q'(x) + 2Q(x)] = 0$$

$$\therefore f(\alpha) = f'(\alpha) = 0$$

(ii) $f(x) = 2x^3 - 5x^2 - 4x + 12$

$$f'(x) = 6x^2 - 10x - 4$$

$$= 2(3x^2 - 5x - 2)$$

$$= 2(x-2)(3x+1)$$

Double root is a solution to $f'(x)=0$

$$x = 2 \text{ or } x = -1/3$$

$$f(-1/3) \neq 0 \therefore x = 2 \text{ is double root}$$

$$\text{Prod. Roots} = -d/a = -6$$

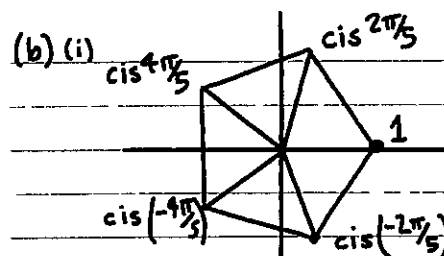
$$\Rightarrow \text{third root} = \frac{-6}{2 \times 2} = -\frac{3}{2}$$

$$\therefore \text{Roots: } 2, 2, -3/2$$

What else would you do in a part ii, except use part i ???

I was surprised how many started with 'let the roots be α, α, β '

Q5 cont.



✓ working

One solution of $z^5 = 1$ is $z = 1$
& the other four solutions are evenly spaced around the unit circle

$$\text{Solutions: } z = 1, \text{cis}(2\pi/5), \text{cis}(4\pi/5), \text{cis}(6\pi/5), \text{cis}(8\pi/5)$$

✓

(ii) Sum roots = $-b/a$

$$1 + \text{cis}(2\pi/5) + \text{cis}(4\pi/5) + \text{cis}(6\pi/5) + \text{cis}(8\pi/5) = 0$$

✓

Reas 4 (ii & iii)

$$1 + (\cos(2\pi/5) + i\sin(2\pi/5)) + (\cos(4\pi/5) - i\sin(2\pi/5))$$

$$+ (\cos(6\pi/5) + i\sin(4\pi/5)) + (\cos(8\pi/5) - i\sin(4\pi/5)) = 0$$

$$\Rightarrow 1 + 2\cos(2\pi/5) + 2\cos(4\pi/5) = 0$$

$$\cos(2\pi/5) + \cos(4\pi/5) = -1/2$$

✓

Parts ii & iii were poorly done. But really, it should have been quite clear on how you should have proceeded.

(iii) $\cos(2\pi/5) + \cos(4\pi/5) = -1/2$

$$\cos(2\pi/5) + (2\cos^2(2\pi/5) - 1) = -1/2$$

$$4\cos^2(2\pi/5) + 2\cos(2\pi/5) - 1 = 0$$

✓

$$\cos(2\pi/5) = \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-1 \pm \sqrt{5}}{4}$$

$$\cos(2\pi/5) = \frac{-1 + \sqrt{5}}{4} \text{ since } 2\pi/5 \text{ is acute}$$

$$\text{ \& so } \cos(2\pi/5) > 0$$

✓

Q5 cont.

(c) (i) $\begin{matrix} \text{1 choice for ref} \\ \text{ways to choose} \\ \text{the 2nd team} \\ \text{of 5 from 6 left} \\ \text{ways to choose} \\ \text{the first team} \\ \text{of 5 from 11} \end{matrix}$

$$\frac{\binom{11}{5} \times \binom{6}{5} \times \binom{1}{1}}{2} = 1386$$

overcounted by a factor of 2!

(ii) Both fussy players play

$$\rightarrow \binom{9}{4} \times \binom{5}{4} \times \binom{1}{1} = 630 \checkmark$$

One of fussy players is a ref

$$\rightarrow \frac{\binom{2}{1} \times \binom{10}{5} \times \binom{5}{5}}{2} = 252 \checkmark$$

Total # ways = $630 + 252 = 882$

Reas 4

Note: we divide by 2 because the two teams of 5 playing each other are equivalent

Some really good attempts but unfortunately not many marks scored in this Q

Question 6 (15 marks)

Calc /9, Reas /6

(a) Required To Prove: $a_n = \frac{n+1}{n!}$

Reas 3

When $n=1$: $\frac{1+1}{1!} = 2 = a_1$

When $n=2$: $\frac{2+1}{2!} = \frac{3}{2} = a_2$

The statement is true for $n=1$ & $n=2$ & so let $k-1$ & k be integers for which the statement is true

ie. $a_{k-1} = \frac{k}{(k-1)!}$ & $a_k = \frac{k+1}{k!}$

Then

$$\begin{aligned} (k+1)a_{k+1} &= a_{k-1} - (k-2)a_k \\ &= \frac{k}{(k-1)!} - \frac{(k-2)(k+1)}{k!} \end{aligned}$$

(USING ASSUMPTION)

$$= \frac{k^2 - (k^2 - k - 2)}{k!}$$

$$= \frac{k+2}{k!}$$

$$a_{k+1} = \frac{k+2}{(k+1)!}$$

thus the statement is true for the next integer, $k+1$.

Hence, by strong induction the statement is true for integers $n \geq 1$

This is an absolutely straight forward induction. Too many gave up on this question too early!

Q6 cont.

(b)(i) $y = 8x^2 - x^4$

$$x^4 - 8x^2 = -y$$

$$(x^2 - 4)^2 = 16 - y$$

$$x^2 = \pm \sqrt{16 - y} + 4$$

$$x = \pm \sqrt{\pm \sqrt{16 - y} + 4}$$

let x_1 & x_2 be the end points of l with $0 \leq x_1 \leq x_2$, then

$$x_2 = +\sqrt{\pm \sqrt{16 - y} + 4}$$

$$x_1 = +\sqrt{-\sqrt{16 - y} + 4}$$

$$\text{Area} = \pi(x_2^2 - x_1^2)$$

$$= \pi((\sqrt{16 - y} + 4) - (-\sqrt{16 - y} + 4))$$

$$= 2\pi\sqrt{16 - y}$$

(ii) Volume = $\int_0^{16} 2\pi\sqrt{16 - y} dy$

$$= \left[\frac{-4\pi(16 - y)^{3/2}}{3} \right]_0^{16}$$

$$= 0 - \frac{-4\pi \times 4^3}{3}$$

$$= \frac{4^4\pi}{3} \text{ units}^3$$

Reas 3

Poorly done!

So many didn't even know where to start.

Calc 2

An easy integral that could be done without (i).

Unfortunately some silly mistakes & some fudging going on.

Q6 cont.

(c)(i) $u = x \cos^{n-1} x$

$$du = x(n-1)\cos^{n-2} x \cdot -\sin x + \cos^{n-1} x$$

$$= -(n-1)x \sin x \cos^{n-2} x + \cos^{n-1} x$$

(ii) $I_n = \int_0^{\pi/2} \underbrace{x \cos^{n-1} x}_u \cdot \underbrace{\cos x dx}_{dv}$

$$= \left[x \cos^{n-1} x \cdot \sin x \right]_0^{\pi/2}$$

$$- \int_0^{\pi/2} \sin x \left(-(n-1)x \sin x \cos^{n-2} x + \cos^{n-1} x \right) dx$$

$$= [0] + (n-1) \int_0^{\pi/2} x \sin^2 x \cos^{n-2} x dx$$

$$- \int_0^{\pi/2} \sin x \cos^{n-1} x dx$$

$$= (n-1) \int_0^{\pi/2} x \cos^{n-2} x - x \cos^n x dx$$

$$+ \left[\frac{1}{n} \cos^n x \right]_0^{\pi/2}$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n - \frac{1}{n}$$

$$n I_n = (n-1) I_{n-2} - \frac{1}{n}$$

$$I_n = \frac{-1}{n^2} + \frac{n-1}{n} I_{n-2}$$

(iii) $I_0 = \int_0^{\pi/2} x dx = \left[\frac{x^2}{2} \right]_0^{\pi/2} = \frac{\pi^2}{8}$

$$I_4 = \frac{-1}{16} + \frac{3}{4} I_2$$

$$= \frac{-1}{16} + \frac{3}{4} \left(\frac{-1}{4} + \frac{1}{2} I_0 \right)$$

$$= \frac{-1}{4} + \frac{3\pi^2}{64}$$

Calc 7

I actually just can't believe how many people couldn't take a hint in (i)!

Also, particularly in recurrence questions like this, you need to concentrate really hard and be SO careful not to make algebraic errors.

Question 7 (15 marks)

Reas /15

$$(a) (i) \quad z = \cos \theta + i \sin \theta \quad (1)$$

$$\frac{1}{z} = z^{-1} = \cos(-\theta) + i \sin(-\theta) \quad (2)$$

$$= \cos \theta - i \sin \theta$$

(by de Moivre)

$$(1) + (2) \Rightarrow z + \frac{1}{z} = 2 \cos \theta$$

$$z = \cos \theta + i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta \quad (3)$$

(by de Moivre)

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta) \quad (4)$$

$$= \cos n\theta - i \sin n\theta$$

(by de Moivre)

$$(3) + (4) \Rightarrow z^n + \frac{1}{z^n} = 2 \cos n\theta$$

$$(ii) \left(z + \frac{1}{z}\right)^5 = z^5 + 5z^3 + 10z + 10 \cdot \frac{1}{z} + 5 \cdot \frac{1}{z^3} + \frac{1}{z^5}$$

$$(2 \cos \theta)^5 = \left(z^5 + \frac{1}{z^5}\right) + 5 \left(z^3 + \frac{1}{z^3}\right) + 10 \left(z + \frac{1}{z}\right)$$

$$32 \cos^5 \theta = 2 \cos 5\theta + 5 \cdot 2 \cos 3\theta + 10 \cdot 2 \cos \theta$$

$$\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$$

Follow the lead in part (i). You don't want to start with the expansion of $(c + is)^5$ — this will give a nice expression for $\cos 5\theta$, not $\cos^5 \theta$

$$(iii) \quad 16 \cos^5 \theta = 15 \cos 3\theta + \cos 5\theta$$

$$\cancel{\cos 5\theta} + 5 \cos 3\theta + 10 \cos \theta = 15 \cos 3\theta + \cancel{\cos 5\theta}$$

$$10 \cos \theta = 10 \cos 3\theta$$

$$\cos \theta = \cos 3\theta$$

$$\theta = 3\theta + 2\pi n, \quad \theta = -3\theta + 2\pi n$$

$$-2\theta = 2\pi n, \quad 4\theta = 2\pi n$$

$$\theta = -\pi n, \quad \theta = \frac{\pi n}{2} \text{ for } n \in \mathbb{Z}$$

This should have been doable without having done i & ii!

[Note: together this is simply $\theta = \frac{\pi n}{2}, n \in \mathbb{Z}$]

Q7 cont.

$$(b) (i) \quad \frac{\text{Area } \triangle RQM}{\text{Area } \triangle RMS} = \frac{\frac{1}{2} \times QM \times h}{\frac{1}{2} \times MS \times h}$$

$$= \frac{QM}{MS}$$

where h is $\perp$ distance from R to line QMS

$$(ii) (a) \quad \frac{BX}{XC} = \frac{\text{Area } \triangle ABX}{\text{Area } \triangle ACX}$$

$$= \frac{\text{Area } \triangle DBX}{\text{Area } \triangle DCX}$$

$$= \frac{\text{Area } \triangle ABX - \text{Area } \triangle DBX}{\text{Area } \triangle ACX - \text{Area } \triangle DCX}$$

$$= \frac{\text{Area } \triangle ABD}{\text{Area } \triangle ACD}$$

A pity not many attempted this question because it's so nice.

This question had nothing to do with similar $\triangle$ s

$$(b) \text{ Similarly } \frac{CY}{YA} = \frac{\text{Area } \triangle BCD}{\text{Area } \triangle BAD}$$

$$\frac{AZ}{ZB} = \frac{\text{Area } \triangle CAD}{\text{Area } \triangle CBD}$$

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB}$$

$$= \frac{\text{Area } \triangle ABD}{\text{Area } \triangle ACD} \times \frac{\text{Area } \triangle BCD}{\text{Area } \triangle BAD} \times \frac{\text{Area } \triangle CAD}{\text{Area } \triangle CBD}$$

$$= 1$$

Q7, cont:

$$\begin{aligned} \text{(c) } a &= \frac{\cos x + \cos y + \cos z}{\cos(x+y+z)} \\ &= \frac{\sin x + \sin y + \sin z}{\sin(x+y+z)} \\ &= \frac{i(\sin x + \sin y + \sin z)}{i(\sin(x+y+z))} \\ &= \frac{\cos x + \cos y + \cos z + i(\sin x + \sin y + \sin z)}{\cos(x+y+z) + i(\sin(x+y+z))} \\ &= \frac{\operatorname{cis} x + \operatorname{cis} y + \operatorname{cis} z}{\operatorname{cis}(x+y+z)} \end{aligned}$$

Also a really nice question
lots of people didn't
attempt

USING THE IDENTITY

$$\begin{aligned} \text{(ii) } a &= \frac{\text{cis } x + \text{cis } y + \text{cis } z}{\text{cis } (x + y + z)} \\ &= \text{cis } (x - (x + y + z)) + \text{cis } (y - (x + y + z)) \\ &\quad + \text{cis } (z - (x + y + z)) \\ &= \text{cis } (-y - z) + \text{cis } (-x - z) + \text{cis } (-x - y) \end{aligned}$$

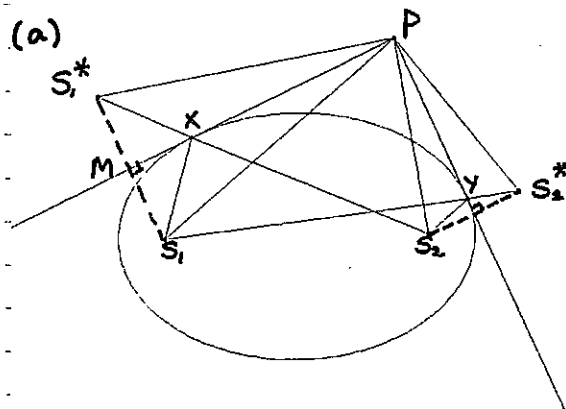
Taking the real part of both sides
(& given $a \in \mathbb{R}$)

$$a = \cos(-y-z) + \cos(-x-z) + \cos(-x-y)$$

$$a = \cos(y+z) + \cos(x+z) + \cos(x+y)$$

Question 8 (15 marks)

Reas /15



There's lots of words here you should have read really carefully in reading time. It would have been impossible to do this question in a rush at the 2½ hour mark if you weren't familiar with the diagram already.

(i) In $\triangle MXS$, & $\triangle MXS^*$

MX (common)

$$\angle S, MX = \angle S,^* MX = 90^\circ \text{ \& } MS, = MS,^*$$

(given S_i^* is a reflection of S_i .)

$$\therefore \triangle MXS_1 \cong \triangle MXS_2 \text{ (SAS)}$$

$$\therefore \angle S, XM = \angle S,^* XM$$

(matching angles in congruent Δ s =) ✓

$$\angle S, XM = \angle S, XP$$

(using property ②) ✓

$$\Rightarrow L_{S,*}XM = L_{S,XP}$$

∴ Since MXP is a straight line
& opposite angles are equal,
 S_1XS_2 is a straight line.

Note: similarly $S_1 Y S_2^*$ is a straight line

Q8 cont.

$$\begin{aligned}
 \text{(ii)} \quad S_1^* S_2 &= S_1^* X + X S_2 \quad (\text{since } S_1^* X S_2 \text{ is a straight line}) \\
 &= S_1 X + X S_2 \quad (\text{matching sides } S_1 X = S_1^* X \text{ in} \\
 &\quad \text{congruent } \Delta S_1 M X \text{ \& } \Delta S_1^* M X) \\
 &= S_1 Y + Y S_2 \quad (\text{using property ①}) \\
 &= S_1 Y + Y S_2^* \quad (\text{matching sides } S_2 Y = S_2^* Y \text{ in} \\
 &\quad \text{congruent } \Delta S_2 N Y \text{ \& } \Delta S_2^* N Y) \\
 &= S_1 S_2^* \quad \checkmark \quad (\text{since } S_1 Y S_2^* \text{ is a straight line}) \quad \checkmark
 \end{aligned}$$

(iii) SSS $\checkmark$

$$\begin{aligned}
 &[\text{Proof: In } \Delta S_1^* P S_2 \text{ \& } \Delta S_1 P S_2^* \\
 &\quad S_1^* S_2 = S_1 S_2^* \quad (\text{from part (ii)}) \\
 &\quad S_1^* P = S_1 P \quad (\text{matching sides = in congruent } \Delta S_1^* M P \text{ \& } \Delta S_1 M P \text{ (SAS)}) \\
 &\quad S_2 P = S_2^* P \quad (\text{matching sides = in congruent } \Delta S_2^* N P \text{ \& } \Delta S_2 N P \text{ (SAS)}) \\
 &\therefore \Delta S_1^* P S_2 \equiv \Delta S_1 P S_2^* \quad (\text{SSS})
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \angle S_1^* P S_2 &= \angle S_1 P S_2^* \quad \checkmark \quad (\text{matching angles in congruent } \Delta S =) \\
 \angle S_1^* P S_2 - \angle S_1 P S_2 &= \angle S_1 P S_2^* - \angle S_1 P S_2 \\
 \angle S_1^* P S_1 &= \angle S_2 P S_2^* \\
 \Rightarrow \angle S_1 P X &= \angle S_2 P Y \quad (\text{since matching angles } \angle S_1 P X \text{ \& } \angle S_1^* P X \\
 &\quad \checkmark \text{ and } \angle S_2 P Y \text{ \& } \angle S_2^* P Y \text{ are equal} \\
 &\quad \text{in congruent triangles})
 \end{aligned}$$

Q8 cont.

$$\begin{aligned}
 \text{(b) (i)} \quad f(x) &= \log_e x - x + 1 \\
 f'(x) &= \frac{1}{x} - 1 \\
 f''(x) &= -\frac{1}{x^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{For max/min } f'(x) &= 0 \\
 \Rightarrow x &= 1
 \end{aligned}$$

$$f''(1) = -1 < 0$$

$\therefore$ maximum occurs at $x = 1$ $\checkmark$

$$\begin{aligned}
 \text{(ii) Maximum value} &= f(1) \\
 &= \log_e 1 - 1 + 1 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore f(x) &\leq 0 \quad \text{for } x \text{ in the domain} \\
 \Rightarrow \log_e x - x + 1 &\leq 0 \quad \text{for } x > 0 \\
 \Rightarrow \log_e x &\leq x - 1 \quad \text{for } x > 0 \quad \checkmark
 \end{aligned}$$

$$\text{(iii)} \quad \log_e(np_1) + \log_e(np_2) + \dots + \log_e(np_n)$$

$$\leq (np_1 - 1) + (np_2 - 1) + \dots + (np_n - 1) \quad \checkmark$$

$$\begin{aligned}
 &= n(p_1 + p_2 + \dots + p_n) - n \\
 &= n \times 1 - n \\
 &= 0
 \end{aligned}$$

$$\therefore \log_e(np_1) + \log_e(np_2) + \dots + \log_e(np_n) \leq 0 \quad \checkmark$$

$$\text{(iv)} \quad \log_e(np_1) + \log_e(np_2) + \dots + \log_e(np_n) \leq 0$$

$$\begin{aligned}
 \log_e(n^{\wedge} p_1 p_2 \dots p_n) &\leq 0 \\
 n^{\wedge} p_1 p_2 \dots p_n &\leq 1 \quad \checkmark
 \end{aligned}$$

This question was much easier to do in a rush than part (a)

$$(v) \quad p_1 p_2 \dots p_n \leq 1$$

$$n^n \frac{x_1}{A} \cdot \frac{x_2}{A} \dots \frac{x_n}{A} \leq 1$$

$$\frac{n^n}{A^n} x_1 x_2 \dots x_n \leq 1$$

$$x_1 x_2 \dots x_n \leq \frac{A^n}{n^n}$$

$$\sqrt[n]{x_1 x_2 \dots x_n} \leq \frac{A}{n}$$

$$\sqrt[n]{x_1 x_2 \dots x_n} \leq \frac{x_1 + x_2 + \dots + x_n}{n} \quad \checkmark$$

$$(vi) \quad \frac{a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd}{10} \geq \sqrt[10]{a^5 b^5 c^5 d^5}$$

$$a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd \geq 10 \quad \text{since } abcd = 1 \quad \checkmark$$

Some tried to do (vi) just using pure inequalities without even thinking to use (v). It's a good idea to just try & get the last mark - but do it sensibly!